Linux Basic Commands

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Source: Linux Basic Commands for MLOps Coursework, Dr. Santosh Kumar

1. File & Directory Navigation

Command	What it does	Example
`pwd`	Print the absolute path of the current directory. Check it before running scripts with relative paths.	`pwd`
`ls`	List files and folders. Flags: -l long, -a hidden, -h human sizes, -t sort by time.	`ls -lah /home/user/project` `ls -lt models/ # newest first`

`cd`	Change directory. .. = up one level, ~ or no argument = home, - = previous directory.	`cd /mnt/data/datasets`
		`cd ..`
		`cd ~/mlops-project`

`find`	Recursively search by name, type, size or modification time.	`find . -name "*.pkl"`
		`find /data -type f -size +500M`
		`find . -mtime -1 # modified in last day`

`which / whereis`	which = path of the executable that would run (handy for checking the active python/pip). whereis also shows man pages and source.	`which python3` `which pip`

2. File & Directory Management

Command	What it does	Example
`mkdir`	Create a directory. -p makes nested directories and avoids an error if it exists.	`mkdir models` `mkdir -p experiments/run_01/checkpoints`
`touch`	Create an empty file, or update its timestamp if it exists.	`touch requirements.txt` `touch src/__init__.py`
`cp`	Copy files or directories. -r recursive, -v verbose.	`cp config.yaml config_backup.yaml` `cp -r dataset/ /mnt/backup/dataset/`
`mv`	Move a file/directory, or rename it if the destination is in the same directory.	`mv model_v1.pt model_v2.pt` `mv results.csv reports/results_2026.csv`

`rm`	Delete files. -r for directories, -f to force. Irreversible, so double-check the path.	`rm old_log.txt`
		`rm -r __pycache__/`
		`rm -rf tmp_checkpoints/`

Note: rm -rf permanently deletes data with no recycle bin. Verify the path first (e.g. with ls), especially in automated scripts.

3. Viewing and Editing File Contents

Command	What it does	Example
`cat`	Print a whole file; also combines files.	`cat config.yaml` `cat file1.txt file2.txt > combined.txt`
`less / more`	Page through large files. Arrows to move, /pattern to search, q to quit.	`less training.log`
`head / tail`	First / last lines (default 10). tail -f streams new lines live.	`head -n 20 data.csv` `tail -f training.log`
`nano / vim`	Terminal text editors. nano is beginner-friendly, vim is more powerful.	`nano train_config.yaml` `vim Dockerfile`

Command	What it does	Example
`grep`	Search text inside files. -i ignore case, -r recursive, -n line numbers.	`grep "ERROR" training.log` `grep -rn "learning_rate" ./src`

4. File Permissions & Ownership

Command	What it does	Example

`chmod`	Set read (r), write (w), execute (x) for owner, group, others. Numeric (755) or symbolic (u+x).	`chmod +x run_training.sh`
		`chmod 644 config.yaml`
		`./run_training.sh`

`chown`	Change owner (and group). Common fix when Docker creates root-owned files.	`sudo chown user:user model.pt` `sudo chown -R user:user ./outputs`
`sudo`	Run one command as administrator (root).	`sudo apt update` `sudo systemctl restart docker`

5. Process & Job Management

Command	What it does	Example
`ps`	List running processes. ps aux shows all, with CPU/memory.	`ps aux | grep python`
`top / htop`	Live process monitor (htop is nicer, may need installing). q to exit.	`top` `htop`
`kill / kill -9`	Stop a process by PID. kill = graceful, kill -9 = force.	`kill 12345` `kill -9 12345`
`& and nohup`	& runs in background; nohup keeps it alive after logout/SSH close.	`python train.py &` `nohup python train.py > train.log 2>&1 &`
`jobs / fg / bg`	jobs lists background jobs, fg brings one to foreground, bg resumes a paused one.	`jobs` `fg %1`

6. System Monitoring & Resource Usage

Command	What it does	Example
`df`	Free/used disk space per filesystem. -h = human-readable.	`df -h`
`du`	Space used by a file/directory. -sh gives a summary; pipe to sort for biggest folders.	`du -sh datasets/` `du -h --max-depth=1 . | sort -h`
`free`	RAM and swap usage. Useful for out-of-memory errors.	`free -h`
`nvidia-smi`	GPU utilisation, VRAM, temperature and processes on NVIDIA GPUs.	`nvidia-smi` `watch -n 1 nvidia-smi`

7. Disk Usage & Storage

Command	What it does	Example
`df -h and du -sh`	First stop when a job fails for lack of disk space: df -h shows free space, du -sh * shows which folders are biggest.	`df -h` `du -sh *`
`watch`	Re-run a command at a fixed interval (disk space, GPU, log folder).	`watch -n 5 df -h`

Tip: run man <command> (e.g. man grep) for the full manual of any command.